ALLO INNOWARE -- GOOGLE SHEETS SYSTEM BLUEPRINT

Detailed Explanation of the 6 Operational Sheets, Data Flow & Separation of Concerns

[*] Tab Name: 'ARCHITECTURAL PHILOSOPHY' | Role: System Foundation

Audience / Primary Actor: Engineering & Management

Summary: CORE DESIGN PRINCIPLE: SEPARATION OF CONCERNS In growing operations teams, spreadsheets fail because companies cram data storage, employee data entry, and management reporting into a single sheet. This causes accidental formula deletion, multi-user concurrency locks, and dirty data. Our prototype strictly organizes operations into 6 purpose-built sheets: Configuration, Catalog, Live Queue, Single-Point UI, Executive KPI Dashboard, and Immutable Audit Log.

Key Columns / Data: Config, Task_Master, Active_Tasks, Employee_Workbench, Manager_Dashboard, Audit_Log

Operational Value: Eliminates spreadsheet chaos by ensuring employees never touch the master database.

1. TAB-BY-TAB DETAILED BREAKDOWN (SHEETS 1 TO 3)

The engine, configuration, and catalog layers

1. Tab Name: 'Config' | Role: System Control Center & Dynamic Parameters

Audience / Primary Actor: Operations Admin / Manager (Set once during onboarding)

Summary: Acts as the master settings page for the entire automation. Centralizes email recipients, daily/periodic cutoff times, grace periods, SLA escalation hours, and the deployed Web App URL.

Key Columns / Data: Key (Col A): MANAGER_EMAIL, LEADERSHIP_EMAIL, DAILY_TASK_CUTOFF_TIME (11:00), DAILY_TASK_MANAGER_ALERT_DELAY_MINS (30), DAILY_TASK_LEADERSHIP_ESCALATION_HOURS (4), WEB_APP_URL, TIMEZONE (Asia/Kolkata). Value (Col B), Description (Col C).

Operational Value: Zero-code maintainability: If management emails or SLA hours change, update cell B4 without modifying script code.

2. Tab Name: 'Task_Master' | Role: Master Recurring Task Catalog & Rules Library

Audience / Primary Actor: Operations Manager / Admin (Updated when new recurring routines are added)

Summary: The permanent blueprint catalog defining all recurring processes across Allo Innoware. Stores frequency rules, assigned owners, working channels, cutoff times, and proof type criteria.

Key Columns / Data: Task_ID (TSK-001..004), Task_Name (Daily Ops Update, Portal Tasks A/B/C), Assigned_To (Employee A..D), Employee_Email, Frequency (Daily, 7 Days, 15 Days, 30 Days), Frequency_Days, Expected_Cutoff_Time, Proof_Type, Target_Portal_Channel, Active.

Operational Value: Whenever a task is completed, the engine consults this catalog to calculate the next cycle date and frequency offset.

3. Tab Name: 'Active_Tasks' | Role: The Live Operational Queue & Timestamp Ledger

Audience / Primary Actor: The Automation Engine (Apps Script) -- Employees NEVER edit directly

Summary: The running heartbeat of the operations system. Tracks every individual cycle instance (e.g. TSK-001-C01, TSK-001-C02) with exact due dates, real-time statuses, and atomic timestamp tracking for audits and anti-duplication.

Key Columns / Data: Instance_ID, Task_ID, Task_Name, Assigned_To, Frequency, Cycle_Number, Due_Date, Due_Time, Proof_Type, Status (Pending, Manager Alerted, Manager Acknowledged, Escalated, Completed), Proof_Submitted, Completed_At, Manager_Alert_Sent_At, Manager_Acknowledged_At, Leadership_Escalation_Sent_At, Next_Cycle_Generated.

Operational Value: Single source of truth. Contains all state flags preventing duplicate alert emails and duplicate cycle creations.

2. TAB-BY-TAB DETAILED BREAKDOWN (SHEETS 4 TO 6)

The user interface, reporting, and compliance layers

4. Tab Name: 'Employee_Workbench' | Role: Single-Point Action UI for Team Members

Audience / Primary Actor: Operations Employees (Employee A, B, C, D) -- Daily operating home

Summary: A clean, filtered user interface designed specifically for low friction. The employee chooses their name from a dropdown in cell B4, and the table dynamically displays ONLY their pending tasks. They enter proof into Column H and click Submit.

Key Columns / Data: Dropdown Filter (B4), Real-Time Pending Count (B5: =COUNTIF), Filtered Table: Instance_ID (Col A), Task_Name (Col B), Frequency (Col C), Due_Date (Col D), Status (Col F), Proof_Type_Required (Col G), Enter Proof / Ticket ID (Col H), Notes (Col I).

Operational Value: Frictionless execution (<15 seconds). Employees never see raw databases or other departments' tasks.

5. Tab Name: 'Manager_Dashboard' | Role: Executive KPI Summary & Compliance Monitor

Audience / Primary Actor: Operations Managers & Executive Leadership (Real-time monitoring)

Summary: High-level operational cockpit displaying live KPI counter cards and employee-by-employee compliance metrics. Automatically computes metrics using dynamic COUNTIFS formulas linked to Active_Tasks.

Key Columns / Data: Top KPI Cards: Total Active Cycles (=COUNTA), Pending Tasks (=COUNTIF 'Pending'), Manager Alerted (=COUNTIF 'Manager Alerted'), Manager Acknowledged, Escalated to Leadership, Completed Cycles. Employee Compliance Matrix: Total Assigned, Pending, Alerted, Escalated, Completed, and Compliance_Rate % (=Completed / Total).

Operational Value: Provides 5-second situational awareness: managers immediately spot bottlenecks without asking for manual status reports.

6. Tab Name: 'Audit_Log' | Role: Immutable Transaction Ledger (The Black Box)

Audience / Primary Actor: System Auditor, Manager, Leadership (Strictly append-only)

Summary: Permanent, un-editable event history recording every single operational state change across the company. Every completion, cycle advance, alert dispatch, manager acknowledgement, and escalation is timestamped here.

Key Columns / Data: Log_ID (LOG-001..), Timestamp (YYYY-MM-DD HH:MM:SS), Event_Type (SYSTEM_INIT, TASK_COMPLETED, CYCLE_CREATED, MANAGER_ALERT_SENT, MANAGER_ACKNOWLEDGED, LEADERSHIP_ESCALATED), Instance_ID, Task_ID, Actor, Details, Status_Before, Status_After.

Operational Value: Complete enterprise auditability: resolves vendor disputes, proves SLA adherence, and leaves zero room for finger-pointing.

3. SYSTEM INTERACTION & DATA FLOW ARCHITECTURE

How data moves across the 6 sheets during the task lifecycle

Lifecycle Step	Trigger / Action	Source Sheet	Target Sheet	System State Transition
1. Task Initiation	System setup / cycle advance	Task_Master	Active_Tasks	Spawns new Instance (e.g. TSK-002-C02) in 'Pending'.
2. Single-Point View	Employee selects name (B4)	Active_Tasks	Employee_Workbench	Filters and displays only assigned pending tasks.
3. Proof Submission	Enters Ticket ID & clicks Submit	Employee_Workbench	Active_Tasks & Audit_Log	Transitions status from 'Pending' -> 'Completed'.
4. Recurrence Spawn	Triggered by completion	Task_Master	Active_Tasks	Calculates Due Date + Frequency, sets Next_Cycle = Y
5. Overdue Alert	Cron engine scans past cutoff	Config	Active_Tasks & Email	Sends HTML alert to Manager; marks 'Manager Alerted
6. 1-Click Acknowledge	Manager clicks button in email	Web App Webhook	Active_Tasks & Dashboard	Stamps 'Manager Acknowledged'; pauses escalation tim
7. SLA Escalation	4 hours elapse unacknowledged	Config & Active_Tasks	Leadership Email	Escalates to Founders; marks status 'Escalated'.
8. Live KPI Update	Formula recalculation	Active_Tasks	Manager_Dashboard	Real-time KPI cards & Compliance % update instantly.

4. WHY THIS ARCHITECTURE PREVENTS SPREADSHEET COLLAPSE

Comparison of traditional manual sheets vs. Allo Innoware automated prototype

Operational Dimension	Traditional Manual Sheets (Failure Mode)	Allo Innoware 6-Sheet Architecture
Data Entry	Everyone edits the same 20-col master sheet; accidental overwrites	Employee Workbench: isolated front-end; zero master sheet editing.
Recurrence Tracking	Employees manually calculate next due dates; cycle drift occurs.	Automated Engine: closes cycle, adds frequency, skips weekends.
Daily Task Proof	Unchecked boxes or verbal claims with zero proof verification.	WhatsApp Protocol: Timestamp + Summary snippet stored in audit trail.
Overdue Notification	Silent failures until an order is late or vendor complains.	Same-day Manager Alert fired automatically at cutoff + grace period.
Manager Response	Requires logging into sheet, finding row, and changing cell.	1-Click Email Action: acknowledge directly from mobile inbox in 2 sec.
Leadership Visibility	Founders micromanage daily tasks or are blindsided by crises.	2-Tier SLA Shield: leadership only notified on manager SLA breach.
Concurrency / Races	Double submits create duplicate rows and repeated alert emails.	LockService concurrency lock + atomic timestamp guards (0 duplicates).
Historical Audit	No change tracking; impossible to know who modified what.	Immutable Audit_Log: records every actor, before/after status & time.

5. KEY FORMULAS & SCRIPTLOCK LOGIC REFERENCE

Technical implementation details behind the formulas and code

[*] Tab Name: 'TECHNICAL SPECIFICATIONS' | Role: Formulas & Code Logic

Audience / Primary Actor: Engineering Evaluators

Summary: 1. DYNAMIC DASHBOARD AGGREGATIONS: - Pending Count: =COUNTIF(Active_Tasks!L2:L, "Pending") - Escalation Count: =COUNTIF(Active_Tasks!L2:L, "Escalated") - Employee Compliance Rate: =IF(Total>0, TEXT(Completed/Total, "0.0%"), "100%") 2. WORKBENCH CONDITIONAL FILTERING: - Dynamic pending task counter: =COUNTIF(F9:F, "<>Completed") - COUNTBLANK(F9:F) - Google Sheets FILTER/QUERY formula maps Active_Tasks where Assigned_To = Employee_Workbench!B4 and Status <> 'Completed'. 3. GOOGLE APPS SCRIPT CONCURRENCY & IDEMPOTENCY: - LockService.getScriptLock().waitLock(30000): Queues concurrent requests up to 30 seconds to prevent race conditions.
- Atomic Flags: Manager_Alert_Sent_At, Leadership_Escalation_Sent_At, and Next_Cycle_Generated are checked before every write. If already stamped, subsequent cron passes immediately abort, ensuring 100% mathematical idempotency.

Key Columns / Data: COUNTIF, COUNTIFS, FILTER, Apps Script LockService, doGet Webhook Web App

Operational Value: High performance: batch array reading (getValues/setValues) prevents hitting Google API quotas.